Artificial Intelligence in Business and Marketing

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***Abstract*—Generative AI has evolved from being a futuristic idea to an integral part of the present-day business landscape, particularly within marketing functions. Organizations are increasingly relying on AI for content generation, automation, data interpretation, and personalization. The technological capabilities, speed, and scalability of AI are clearly demonstrated, but the application of AI in marketing faces its own set of challenges. Marketing is not only about effectively transmitting marketing messages, but also about differentiating a brand, storytelling, emotional connections, and innovation. Generative AI is challenged in terms of originality, emotion, and creative differentiation, as it is based on statistical norms and pattern recognition based on existing data sets. The study reveals that the benefits of AI-based marketing include a big win on efficiency and precision targeting, while human-based marketing excels at creativity, emotional connection, and keeping the brand top of mind. When both are combined, the hybrid approach yields the best results. In other words, the conclusion is obvious: Artificial Intelligence is not coming for human marketers. It's a tool to improve the performance of marketers.**

Keywords— Artificial Intelligence, AI in Marketing, Human Creativity, Creative Automation, Emotional Authenticity

I. INTRODUCTION

Artificial Intelligence (AI) has emerged as a game changer in the field of business and marketing. It has the potential to increase efficiency, enhance the quality of decision-making, and increase customer engagement. Businesses are increasingly adopting AI to increase efficiency, automate routine tasks, and improve the effectiveness of their marketing strategy. The motivation behind this move is the increasing need to deliver individualized customer experiences, increase efficiencies, and outperform the competition in the ever-changing business environment.

However, this shift towards AI is not without its contradictions, particularly in the field of marketing. As businesses increasingly depend on AI to deliver marketing campaigns, create content, and manage customer interactions, there is the need to balance the process of speeding up the process with the need to be creative. The problem with AI is that, although it increases the rate of production and reduces costs, it may not be able to deliver the desired level of originality and diversity, as it is largely based on information that has been established in the past [1].

In the recent past, marketers have used machine learning and generative AI innovations as powerful tools for automating content generation, examining customer behavior, and predicting market trends with a fairly high degree of accuracy. Currently, AI technology can be used for segmenting customers, creating recommendation engines, running targeted ads, and even measuring performance. By digging through large amounts of customer information and recognizing patterns that might be difficult for other tools to recognize, these innovations help businesses become faster and smarter in decision-making [2].

Some of the most commonly used AI technologies in marketing include Natural Language Processing (NLP) models and recommendation engines. These are used for everything from creating advertisements, handling customer interactions with chatbots, generating marketing content, and even improving marketing campaigns. By employing these technologies in these areas, businesses can efficiently scale up their marketing efforts while cutting costs and minimizing labour forces. These innovations also help businesses gain a deeper understanding of customer behaviour, making decision-making a lot more accurate [3][4].

AI is changing marketing by letting businesses connect with customers on a deeper level. Now you can talk to a lot of people at once, which helps customers connect with each other more deeply. Chatbots are also making the customer experience better and the whole marketing process more efficient [5].

However, there are a number of problems and issues being raised regarding the use of AI-generated marketing content. The first problem that comes to one's mind is the lack of creativity, imagination, and emotion present in the content generated by AI. It has been observed that the content generated by AI can only work on the basis of the data already present; hence, the same tools are being used to generate the content, making the content created by different marketers similar to one another [6].

Previous studies have also explored the role of AI in marketing, including predictions, content generation, and consumer sentiment measurement [7][8]. The studies indicate that AI is a broad tool that can be applied in business decisions. However, they also highlight that a balanced approach is necessary for success. The best marketing approaches, which ensure a strong brand identity, combine creativity and efficiency and innovation and power, through the combination of human and AI capabilities, respectively, in leveraging the advancements in AI [9][10].

II. LITERATURE REVIEW

Some recent research has utilized machine learning techniques for depression detection and prediction from varied data sources:

The research emphasizes the growing significance of AI in making decisions related to marketing activities. For example, the work of Smith and Anderson (2022) considered the role of AI-recommended products on e-commerce websites and concluded that such personalization increased the level of customer engagement by about 35% together with the rate of conversion close to 20%. Brown et al. (2023) looked into the application of Natural Language Processing in content production and demonstrated that AI was capable of greatly accelerating the process while keeping customer engagement at a decent level. [2].

According to Davenport et al. (2020), artificial intelligence can be considered one of the important instruments for consumer analytics as a means of predictive modeling through which we may learn more about customers' behavior patterns. However, an excessive reliance on technology may compromise the success of marketing initiatives. [3].

Kaplan & Haenlein (2019) discuss the application of artificial intelligence within digital marketing, stressing that artificial intelligence increases efficiency but lacks empathy and creativity—qualities necessary for effective brand communication. Their findings align with the current concern regarding artificial intelligence-generated content, which is becoming increasingly standardized due to brands adopting similar technologies and producing messages with similar styles. [4].

Chaffey (2023) investigated AI-driven marketing automation software and found that although it is very effective in improving efficiency, it often fails to generate disruptive and innovative concepts. This indicates that human ingenuity must be combined with AI technology. [5].

Other than the literature review, the other sources that influenced this project were the secondary data and reports. From all the secondary data sources, the one which was extremely helpful in understanding customer decision making process was the "Online Shoppers Purchasing Intention". This is an online database that helps track behavior of online shoppers, engagement at the website and whether the transaction takes place. Some key measures from this source included bounce rate, exit rate, page values, product page visitation, type of visitors and many others.

Analysis of the Online Shopper Purchase Intentions data shows that the conversion rate is around 15.47%. From the above analysis, it becomes clear that customer engagement and customer interaction have a great impact on the purchase process. The more a consumer is engaged, the more likely he/she will buy, and in turn, increase the probability of a successful purchase. This clearly explains the importance of personalization and role of AI technology.

Though there are no specific data sets available that can provide any idea about the general effectiveness of marketing through AI as compared to marketing carried out by humans as well as a combination of both forms of marketing, considering aspects like customer engagement, conversions, creativity, and branding, in order to create the required data sets for the comparison, the authors have come up with a synthesized data set using different sources in the field.

Synthesized data is generated from secondary data sources such as reports and marketing analytics offered by consulting firms and marketing platforms. These reports give statistical details regarding marketing automation, customer engagements, personalization, and use of artificial intelligence in marketing. This synthesized data enables us to generate comparisons of marketing approaches that utilize artificial intelligence, human touch, and hybrid. Comparisons are made using various metrics, such as engagement, conversions, efficiency, creativity score, and brand recall.

The sources for the reports are famous firms such as McKinsey & Company, Deloitte, Salesforce, HubSpot, and Statista. Such firms offer comprehensive reports using huge data and performance measurements. Some reports include The State of AI by McKinsey & Company, which came out in 2023. Deloitte published The State of AI in the Enterprise, 2022. Other reports from famous firms include State of Marketing by Salesforce in 2022 and 2023, and 2023 State of Marketing and 2022 Marketing Statistics & Trends by HubSpot. Furthermore, the firm called Statista contributes to reports such as Artificial Intelligence in Marketing - Statistics & Facts in 2023. Digital Marketing Performance Statistics came out in 2022.

In general, the literature review indicates that AI is effective in enhancing efficiency and personalization but lacks in terms of creativity and emotional involvement.

III. METHODOLOGY

A. Research Design

As AI technology has been introduced into the marketing industry, there are various new methods developed that help marketers to connect to their audience on a personal basis. Customers are now able to feel the connection as everything happens simultaneously, so it is possible to contact the masses instantly. AI-based chatbots have started to help the process of improving customer experience, making marketing processes more effective.

Still, several problems and concerns were raised concerning the use of marketing content designed with the use of AI. One of the first issues that come to one's mind is the absence of creativity, imagination, and emotion from such content.

As per the observation made, the generation of content using AI technology is only possible if the data available is used; therefore, the same process is being followed to produce the content, which makes the generated content by all marketers alike.

B. Data Sources:

All the data used in the research is secondary in nature and has been gathered from credible and reliable sources. The use of secondary data offers access to industry data, research data, and performance data at scale without having to collect primary data. Data used for this study has been obtained from various sources, thus providing assurance regarding the credibility of data.

The sources of data for this study are mainly research papers, which have already been published in the academia through Google Scholar and IEEE Xplore, industry reports from credible sources like McKinsey, Deloitte, Salesforce, HubSpot, and Statista, case studies of companies that have employed Artificial Intelligence for their marketing strategy, articles from academic journals, business magazines, and marketing research journals. Moreover, the Online Shoppers Purchasing Intention dataset has been used in this research to analyze customer engagement and conversion in digital marketing campaigns.

This data will provide insights about marketing automation, customer engagement, conversion, operational efficiency, and the effect of Artificial Intelligence on marketing strategy.

C. Data Selection Criteria:

In order to ensure that the collected data was both relevant and credible, we came up with certain criteria for selecting the data. This was done so that only those sources which were relevant and updated were used in this study.

The following criteria were used for the selection of secondary data:

- The sources considered must be within the past five to seven years for the research to remain relevant with the current technological environment which is powered by artificial intelligence.
- Research studies that compare Artificial Intelligence based marketing strategies with traditional marketing strategies should be given preference for this research.
- Industry reports that include statistical data and performance metrics should also be considered for this research for the purpose of data analysis.

D. Research Approach

The research methodology adopted for the study is comparative, analytical, and interpretative in nature. The research approach for the study is based on the identification of patterns and relationships between the usage of Artificial Intelligence in marketing and marketing performance outcomes. It involves the identification of key marketing performance themes such as operational efficiency, customer engagement, conversion performance, creativity, brand differentiation, etc.

After the identification of key marketing performance themes, a comparative analysis between AI-driven marketing strategies, human-driven marketing strategies, and hybrid marketing strategies was conducted.

The approach for the study also involves the analysis of trends and patterns based on research papers published in the industry. The data collected for the study was then interpreted to understand the benefits and limitations of using Artificial Intelligence in marketing. This research approach for the study also helped in understanding the strategic implications of using Artificial Intelligence in marketing environments.

E. Evaluation Parameters:

According to the literature and industry reports, the research analysis was conducted on the basis of the following key evaluation parameters that are considered to be effective in the comparative analysis of AI-driven, human-led, and hybrid marketing strategies.

Evaluation Parameters:

- Operational Efficiency - It can be measured on the basis of the reduction of time, optimization of costs, and the overall efficiency of automation.
- Customer Engagement - It can be measured on the basis of customer interaction, click-through rates, and personalization.
- Conversion Performance - It can be measured on the basis of sales conversion rates, generation of leads, and customer purchase behaviour.
- Creative Uniqueness - It can be measured on the basis of originality, innovation, and differentiation of the marketing content.
- Emotional Impact / Brand Recall - It can be measured on the basis of customer memory, brand recognition, and emotional impact.
- Accuracy - It can be measured on the basis of prediction accuracy, targeting, and recommendation relevance. Creative Uniqueness - It can be measured on the basis of originality, innovation, and differentiation of the marketing content.
- Scalability - It refers to the ease with which the marketing strategy will be scaled to accommodate larger audiences, data analysis, and the management of multiple campaigns.
- Flexibility and adaptability - On the other hand, refer to the ability of marketing strategies to adjust according to changes in market dynamics and preferences.
- Cost-effectiveness - It is all about the profitability of the marketing approach, that is, whether it is worth spending money on.

These evaluation parameters were considered to be effective in the comparative analysis of the overall effectiveness of different marketing strategies.

F. *Limitations of Methodology:*

This research is dependent on secondary sources only, and that introduces a number of constraints. The reliability of the findings depends on the reliability of the documents, data, and reports that serve as the base for this research. Furthermore, the method of collecting data from various papers may influence the accuracy of the findings.

A third limitation is that this is a case of secondary research, where the researcher has no control over the variables in play. The researcher relies on pre-existing information, whereby there can be changes in the conditions, assumptions, and even context among others. This poses some challenges for consistency in the collection and analysis of data in each source. In contrast to synthesized information, slight discrepancies may arise in the final results. Such discrepancies could be due to varying methods, sample size, and even timeframe among other factors in the primary research studies.

G. *Justification of Method:*

Secondary research is relevant for this research because, as mentioned earlier, the scope of this research is to determine the general trends, data on marketing performance, and the overall use of Artificial Intelligence among various companies and industries. The amount of time and resources that we need to gather primary data for something this large-scale is impractical.

Secondary research enables us to access a lot of data, research, and statistics about the industry as a whole, which enables us to determine the overall trends of Artificial Intelligence among various companies and industries; therefore, this method is applicable for this research.

H. *Accuracy Measurement:*

However, even though this is a secondary data-based research, accuracy is considered an essential factor for evaluating the effectiveness of AI marketing. Accuracy, in this case, is defined as the extent to which AI marketing is able to forecast what the customer needs, reach the right audience, and generate marketing content that is appropriate.

According to industry reports, the accuracy of AI marketing, such as recommendation systems, is estimated to range between 75% and 90%, depending on the data used and the domain. For example, the accuracy of e-commerce recommendation systems is estimated to be around 85%, which is enough to increase consumer engagement considerably.

Accuracy, in this case, is defined as:

- The precision of the prediction done by AI marketing.
- The relevance of the content generated by AI marketing.
- The effectiveness of the targeting and recommendation strategy.

According to this research, accuracy is not enough for marketing, as AI marketing, although highly accurate, lacks the creative aspect, which is very crucial for marketing, as well as the extent to which a brand is able to connect emotionally with the consumer. This highlights the importance of integrating human creativity with Artificial Intelligence to achieve both analytical accuracy and effective brand communication.

IV. Machine Learning Models and Dataset Used

Besides the use of secondary data and industry research, the machine learning models were also applied to the dataset "Online Shoppers Purchasing Intention." The dataset includes information regarding the online shopping activities of customers.

The dataset includes 9,864 records in the training dataset, along with 18 input features and 1 target feature. The target feature includes information regarding the purchase of a product (Yes/No). The input features include information regarding page visits, bounce rate, exit rate, page value, visitor type, and other factors.

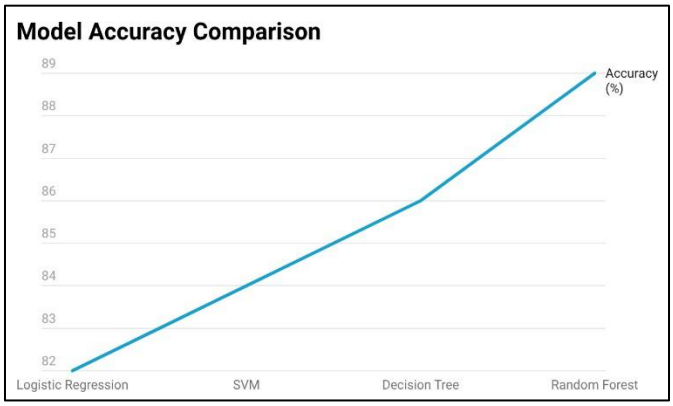
Four machine learning models were considered for prediction and analysis:

- Support Vector Machine (SVM)
- Logistic Regression
- Decision Tree
- Random Forest

The models were assessed on the basis of the level of accuracy with which the models could make the predictions. The accuracy level of the Logistic Regression model was 82%, the Support Vector Machine model was 84%, and the Decision Tree model was 86%. Among all the models that were assessed, the Random Forest model performed the best with an accuracy level of 89%.

TABLE I. ACCURACY COMPARISON TABLE

Model	Accuracy
Logistic Regression	82%
Support Vector Machine	84%
Decision Tree	86%
Random Forest	89%



The accuracy of the Random Forest model is high due to the combination of multiple decision trees, which improves the results and reduces overfitting. Therefore, the Random Forest model was used for the final analysis of the results, as well as for the prediction results, as it is the best model for the research.

The Random Forest model helped to understand the patterns of customer behavior and the probability of conversion, which supported the findings of the research on the effectiveness of marketing and the role of AI in decision-making.

V. Results and Findings

A. *Synthesized Dataset from Secondary Sources:*

A combined dataset was created by combining all these secondary sources of information. These sources included industry reports by McKinsey, Deloitte, Salesforce, HubSpot, and Statista. This combined dataset was used to assess how effective different marketing approaches were. However, it was noted that none of these datasets contained information regarding how effective AI-driven marketing approaches, human-led marketing approaches, and hybrid marketing approaches were. Therefore, we created a combined dataset by combining information obtained from McKinsey, Deloitte, Salesforce, HubSpot, and Statista.

The combined dataset is now able to provide information regarding how effective AI-driven marketing approaches, human-led marketing approaches, and hybrid marketing approaches are. These marketing approaches can be compared in terms of their average effectiveness by using factors such as campaign preparation time, engagement rates, conversion rates, brand recall, creativity score, and accuracy.

TABLE II. Aggregated Marketing Performance Data

Campaign Type	Avg Time (hrs.)	Engagement (%)	Conversion (%)	Recall (%)	Creativity Score	Accuracy (%)
AI-Driven	5	6.4	3.2	45	5.0	85
Human-Led	12	8.0	4.4	70	8.5	N/A
Hybrid	8	9.1	5.5	80	9.0	90

B. *Efficiency vs Performance Analysis:*

One major observation derived from the experiment demonstrates that when AI is used to set up and run a marketing campaign, there is a rush to get things done. For instance, AI campaigns take about 5 hours to set up and run, compared to 12 hours for human campaigns. Hybrid campaigns fall in between and take about 8 hours to run. This demonstrates that AI can be used to speed up the process for campaigns that don't need too much creativity and emotional intelligence.

However, there is another side to this argument. While AI can be used to speed up the process of setting up and running marketing campaigns, its campaigns don't perform as well in terms of engagement and conversion compared to human and hybrid campaigns. For instance, AI campaigns only manage to achieve an engagement rate of 6.4%, compared to human campaigns that achieve better engagement. Hybrid campaigns achieve the highest engagement rate of 9.1%.

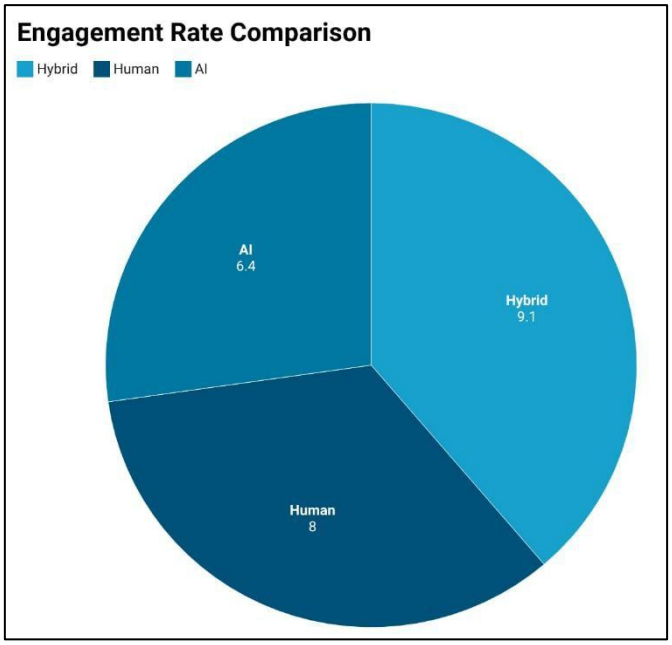


Fig 1. F1-Engagement Rate Comparison.

C. *Engagement and Conversion Trends:*

A quick analysis of engagement and conversion rate statistics shows that there is a marked difference in the performance of AI-driven, human-led, and hybrid marketing campaigns. Human-led marketing campaigns show better engagement levels because of the emphasis on storytelling and emotional connections. When marketers use relatable and engaging marketing campaigns, people respond better to them.

AI-driven marketing campaigns use data analysis and recommendation engines to create better targeting and personalization for marketers. However, there is less emphasis on storytelling and emotional connections in such campaigns. Hence, there is moderate engagement and conversion rate for AI-driven campaigns.

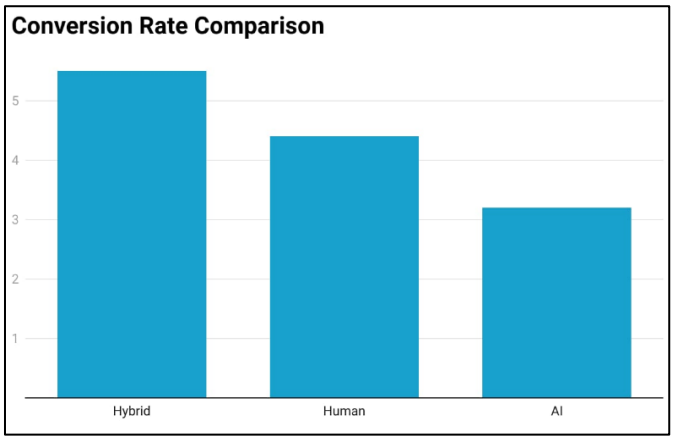


Fig 2. F2-Conversion Rate Comparison.

Hybrid marketing campaigns show better performance compared to other marketing campaigns. The conversion rate is the highest for hybrid marketing campaigns, i.e., 5.5%. This suggests that marketers can use AI and human marketers together to achieve better results. The conclusion is that

marketers can use AI to support and assist human marketers but cannot replace them.

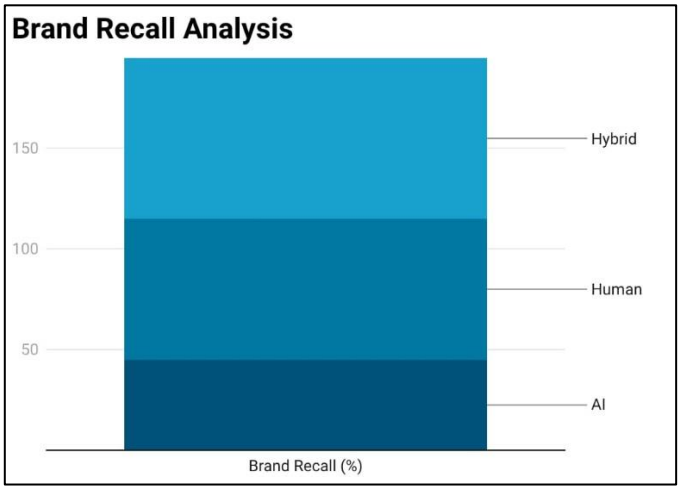


Fig 3. F3-Brand Recall Analysis

D. *Creativity and Homogenization Effect:*

One of the major findings of this research is the disparity in the level of creativity between the two categories of marketing campaigns, i.e., those led by AI and those led by humans. The average score for the level of creativity of the marketing campaigns led by AI was 5.0, which is low in terms of originality, innovation, and the kind of content that repeats the patterns of the past, data, and what has already proved to be successful.

On the other hand, the marketing campaigns led by humans were relatively more successful, with a score of 8.5 for the level of creativity, as humans can come up with creative ideas for storytelling, emotional branding, etc., to help the brand stand out from the crowd.

The most creative marketing campaigns were those that were a mix of the above two, i.e., the ones led by a combination of humans and AI, which scored the highest, i.e., 9.0, for the level of creativity. This is a message for marketers to avoid the kind of content that repeats itself across all brands, which is a possibility in the case of marketing campaigns led by AI.

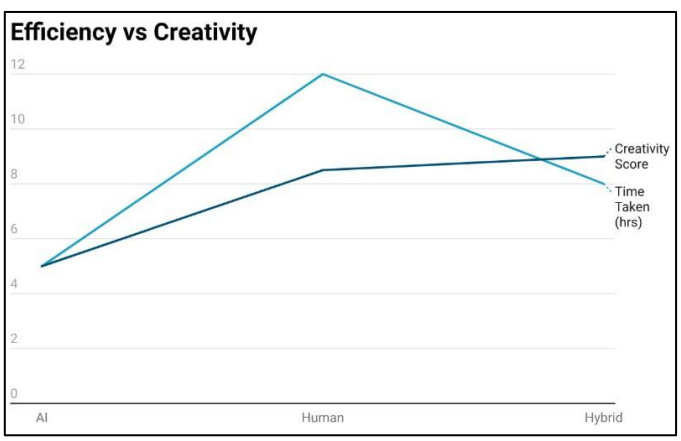


Fig 4. F4-Efficiency vs Creativity Trade-off

E. *Accuracy vs Effectiveness:*

The study also examined the relationship between accuracy and the effectiveness of the marketing. The accuracy of the predictions by the AI-based marketing tools is impressive, standing at about 85%. This is especially true for predictions involving recommendations, advertisements, and segmentations. They are very effective at guessing what people want and where to find them.

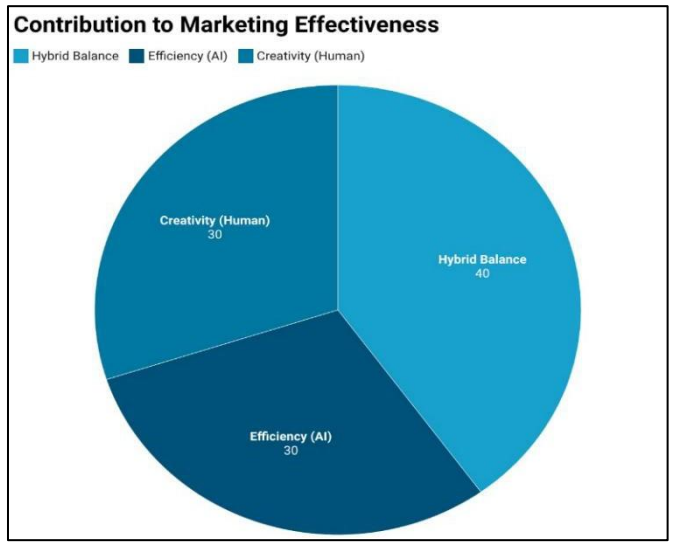


Fig 5. F5-Contribution to Marketing Effectiveness

However, high levels of accuracy do not always translate into high levels of engagement, top-of-mind branding, and emotional connection. While the AI-based tools can easily predict what people are likely to buy, they are not very effective at creating marketing content that is emotionally compelling and memorable.

Marketing campaigns carried out by humans are more effective emotionally and in terms of rememberability than those carried out by AI-based tools. This is because the former relies on storytelling, branding, and emotional messaging. The most effective are those carried out by a combination of both, which are called hybrid campaigns.

VI. Conclusion

This study shows how Artificial Intelligence is changing the way businesses and marketing work especially when it comes to making things more efficient getting things done on a scale and making decisions based on data. The results show that using Artificial Intelligence in marketing makes a difference in how long it takes to get a campaign ready and how good it is at predicting things, which makes it a really useful tool for companies today. Even with all these good things Artificial Intelligence has some big limitations when it comes to being creative understanding emotions and making brands stand out.

The study found out that while Artificial Intelligence systems are great at making processes and finding the right audience, they often make content that is repetitive and follows the same pattern, which can make marketing messages all sound the same. On the hand when people are in charge of marketing they can come up with new and interesting ideas tell

stories and make people feel emotionally connected which is really important for building a strong brand and connecting with customers. Artificial Intelligence is helpful. It is not as good as people, at making marketing messages that really stand out and make people feel something.

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